EXPERIMENT 9: CISCO PACKET TRACER – CLI

AIM:

To configure basic network devices using the Cisco IOS Command-Line Interface in Packet Tracer.

PROCEDURE:

1. Place one router, one switch and two PCs in Packet Tracer.

2. Connect the PCs to the switch and connect the switch to the router using suitable Ethernet cables.

3. Assign IP addresses to the PCs and router interface in the same subnet.

4. Open the router CLI, enter privileged EXEC and global configuration modes.

5. Configure the router interface, enable it and verify the configuration with show commands.

6. Ping the router interface from the PCs.

7. Use the correct interface name for the router model selected in Packet Tracer.
8. Verify that the router interface status changes to up/up after the cable is connected and no shutdown is applied.
9. Save the configuration if required and check the routing and interface information.
10. Use Simulation mode to observe packet movement if additional verification is needed.

CODE:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# end
```

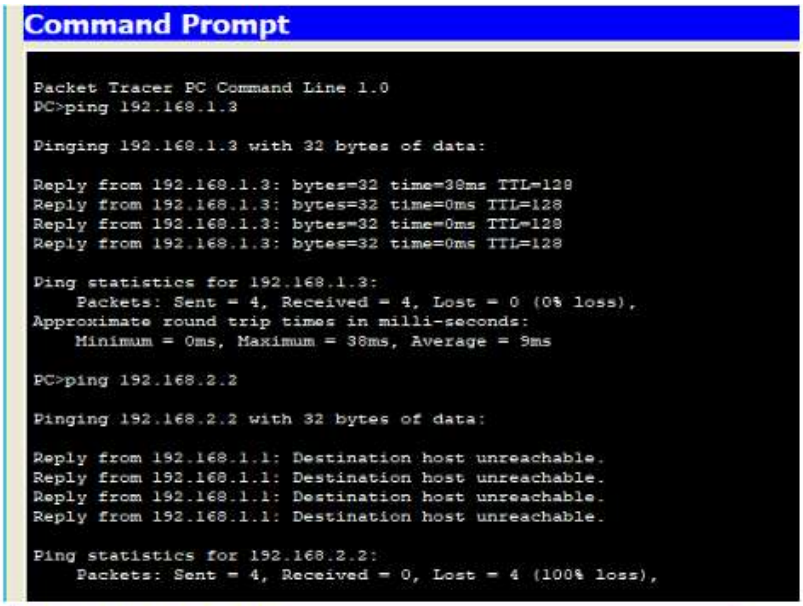
```
Router# show ip interface brief
Router# show running-config

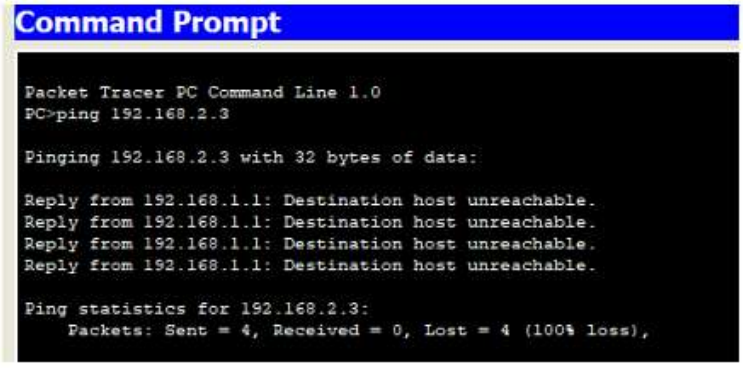
PC1:
IP address: 192.168.1.10
Subnet mask: 255.255.255.0
Default gateway: 192.168.1.1

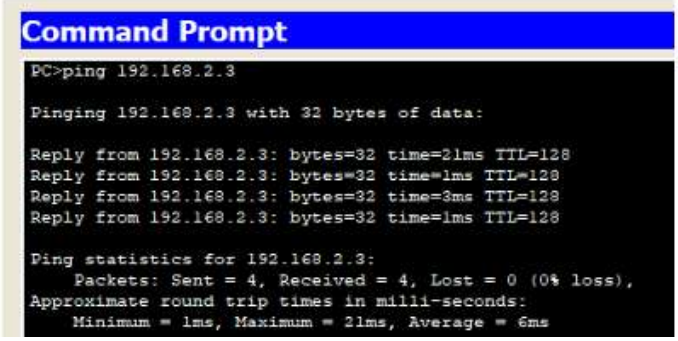
PC2:
IP address: 192.168.1.11
Subnet mask: 255.255.255.0
Default gateway: 192.168.1.1

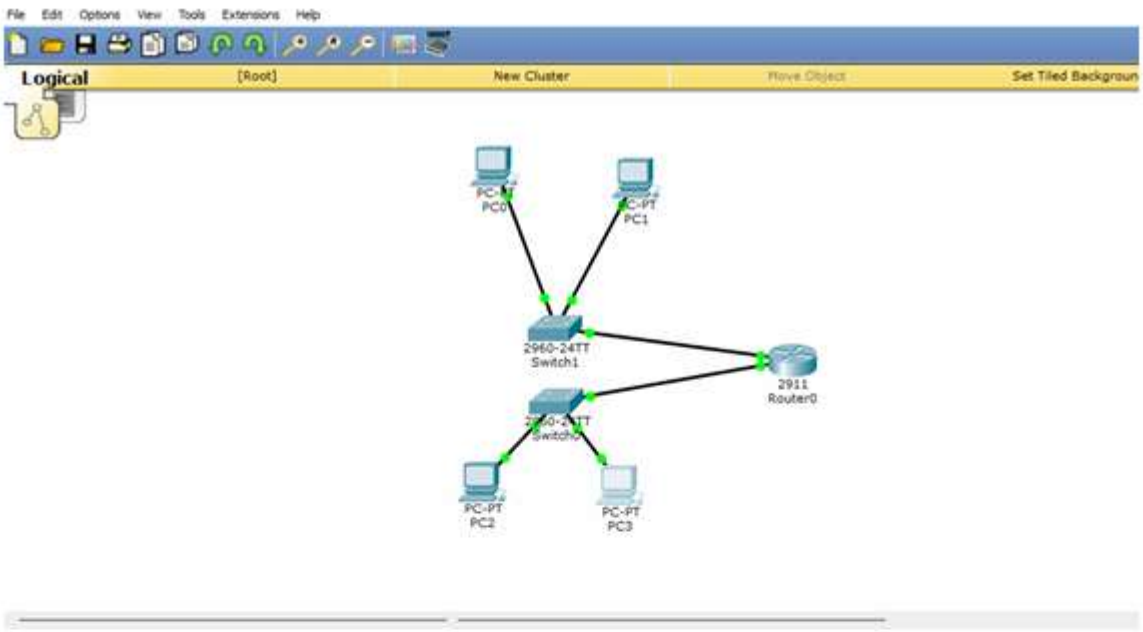
PC command:
ping 192.168.1.1
```

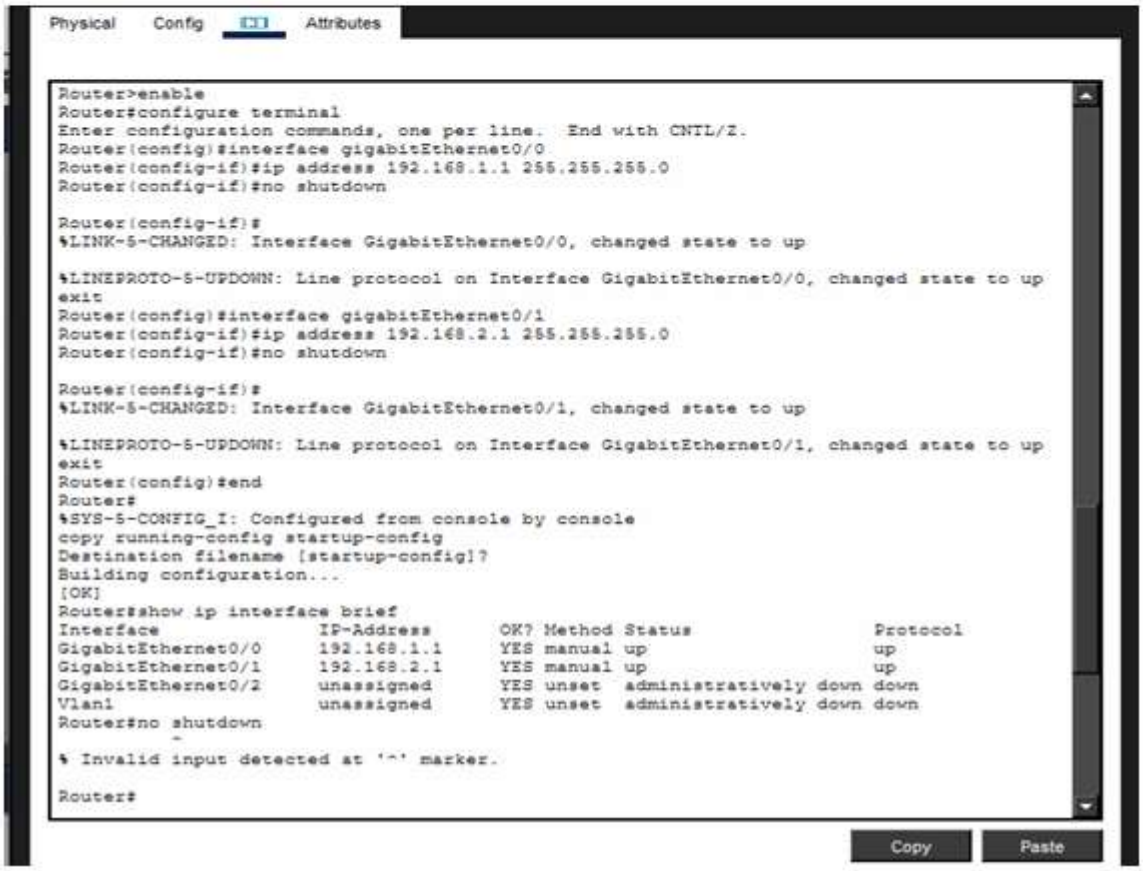
OUTPUT:


Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=30ms TTL=128
Reply from 192.168.1.3: bytes=32 time=0ms TTL=128
Reply from 192.168.1.3: bytes=32 time=0ms TTL=128
Reply from 192.168.1.3: bytes=32 time=0ms TTL=128
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 38ms, Average = 9ms
PC>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),


Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),


Command Prompt
PC>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time=21ms TTL=128
Reply from 192.168.2.3: bytes=32 time=1ms TTL=128
Reply from 192.168.2.3: bytes=32 time=3ms TTL=128
Reply from 192.168.2.3: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 21ms, Average = 6ms





RESULT:

Thus, basic router interface configuration and verification were performed using Cisco IOS CLI in Packet Tracer.